https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	θ is the independent variable and *t* is the dependent variable **or** vary θ and measure *t*	**1**
	keep *d* constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • plane supported by stand / support • stand/support on bench/floor/horizontal surface • minimum of two labels from cube, cylinder, (inclined) plane, method of support, bench/floor/horizontal surface, pulley, string	**1**
	diagram showing method to measure *d*, e.g. clamped vertical rule near cylinder **or** drawn rule on plane used to measure *d* **or** distance *d* marked on plane and rule used to determine *d*	**1**
	use a protractor to measure *θ* **or** use a rule(r) to measure appropriate lengths for a trigonometric calculation	**1**
	use a timer/stop-watch to measure *t* **or** light gates connected to a timer to measure *t*	**1**

https://xtremepape.rs/

Question	Answer	Marks
1	**Method of Analysis**	
	plot a graph of $\frac{1}{t^2}$ against sin θ or equivalent (e.g. sin θ against $\frac{1}{t^2}$) Do not accept logarithms.	**1**
	$H = -\frac{2d(A+B)\times \text{gradient}}{A}$ (for sin θ against $\frac{1}{t^2}$: $H = -\frac{2d(A+B)}{A\times \text{gradient}}$)	**1**
	$K = -\frac{2d(A+B)\times y\text{-intercept}}{A}$ (for sin θ against $\frac{1}{t^2}$: $K = -H\times y$-intercept)	**1**